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Stock market index prediction based on reservoir computing models

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ABSTRACT

Prediction of the financial market price is critical for financial decision-making and market policy-making. Recently, various machine learning and deep learning methods have been adopted to predict financial markets' movements using historical time series of prices. However, accurate prediction of financial prices is still a long-standing challenge that always calls for new approaches. In this study, a novel machine learning model of reservoir computing is developed to predict stock market indices. The performance of the proposed new model is systematically evaluated using the time series of daily closing prices of seven major international stock market indices including S&P500 Index, New York Stock Exchange Composite, Dow Jones Industrial Average, Nasdaq Composite Index, Financial Times Stock Exchange 100 Index, Nikkei 225 Index, and Shanghai Stock Exchange Index between January 4, 2010, and December 31, 2018 covering 2,272 trading days. The results show that our model outperforms the widely used deep learning methods of long short-term memory and recurrent neural network in most cases. To further evaluate the predictive capability of our model, we compare our model to the other two newly reported deep learning methods in recent studies. Comparative results also show that our model is competitive to those deep learning methods in predicting stock market indices. Our study contributes to the literature by developing novel reservoir computing models for financial market predictions. Meanwhile, our results also provide practical implications for financial practitioners of potential financial applications of reservoir computing in financial time series analysis and predictions.

1. Introduction

Prediction of stock price movements is essential for investment decision-making, portfolio management, and risk management. Though researchers and practitioners have made great efforts to understand the price dynamics (Fama, 1995), and various price prediction methodologies have been developed (Abu-Mostafa & Atiya, 1996; Zhou, min Zhou, Yang, & Yang, 2019; Baek & Kim, 2018; Chong, Han, & Park, 2017; Weng, Lu, Wang, Megahed, & Martinez, 2018), successful stock market prediction remains as one of the most challenging problems in the studies of financial markets (Abu-Mostafa & Atiya, 1996) for the chaotic nonlinearity and volatility (Yu & Yan, 2020; Rezaei, Faaljou, & Mansourfar, 2020).

In traditional financial economics, econometrics provides statistical predictive models such as autoregressive moving average model (ARMA) (Rounaghi & Zadeh, 2016; Rojas et al., 2008) and autoregressive integrated moving average model (ARIMA) (Ariyo, Adewumi, & Ayo, 2014; Babu & Reddy, 2014; Banerjee, 2014). In the past few years, machine learning (ML) and deep learning (DL) have achieved significant developments in various domains (Silver et al., 2016; LeCun, Bengio, & Hinton, 2015). In finance, there is a growing enthusiasm in utilizing techniques of machine learning in prediction of stock price movements such as K-nearest neighbor (Qian & Rasheed, 2007), decision tree (Qian & Rasheed, 2007), support vector machine (Trafalis & Ince, 2000, 2005, 2007, Patel, Shah, Thakkar, & Kotecha, 2015b, 2015a), random forest (Patel et al., 2015b; Patel, Shah, Thakkar, & Kotecha, 2015a; Ballings, den Poel, Hespeels, & Gryp, 2015; Zhang, Cui, Xu, Li, & Li, 2018; Cervelló-Royo & Guijarro, 2020), Naïve-Bayes (Patel et al., 2015a), and

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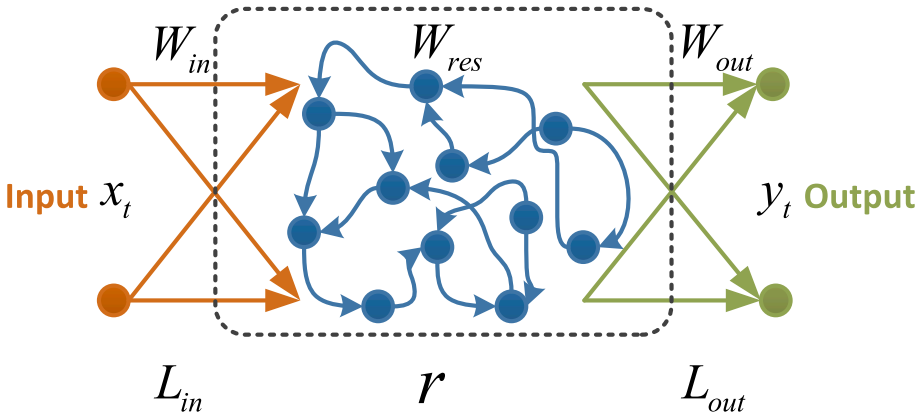


Fig. 1. Diagram of a standard reservoir computing model in which the input vector x_t of input layer L_{in} is projected into the reservoir r and the output vector y_t is readout from the output layer L_{out}. Both of W_{in} and W_{res} are kept un-trained while only W_{out} is trained.

AdaBoost (Ballings et al., 2015). Unlike these statistical learning approaches, a class of deep learning algorithms has been introduced more recently to predict stock prices. A variety of layered neural network structures such as artificial neural network (ANN) (Kim & Lee, 2004; Chang, Liu, Lin, Fan, & Ng, 2009; Dutta, Jha, Laha, & Mohan, 2006; Qian & Rasheed, 2007; Patel et al., 2015a; Zhang & Wu, 2009), recurrent neural network (RNN) (Zhang, Aggarwal, & Qi, 2017), and long short-term memory (LSTM) (Nelson, Pereira, & de Oliveira, 2017; Zhang, Yan, & Aasma, 2020) are proposed and demonstrate various capabilities in the prediction of financial markets. More recently, investigations on the activation functions of DL models reveal that rectified linear unit (ReLU) works better in the prediction of multiple markets and various frequencies (Orimoloye, Sung, Ma, & Johnson, 2020). Rather than directly predicting stock price itself, some studies focus on classifications of the stock market trend using predictive models (Zhang et al., 2018). Most studies target the prediction of daily prices. There are also study reports that novel ML and DL predictive models also work for intraday price time series (Arévalo, García, Guijarro, & Peris, 2017) and challenging long-term market trend prediction (Liu, Gong, Yang, & Chen, 2020; Cervelló-Royo & Guijarro, 2020). Instead of using price time series as direct input to the predictive models, some studies first build traditional technical indicators from the time series and conduct feature selections to reduce dimension before input into deep generative models (Haq, Zeb, Lei, & Zhang, 2021) or hybrid fuzzy neural networks (Garcia, Guijarro, Oliver, & Tamošiūnienė, 2018). In another approach, neural network-based feature extractors are proposed to predict stock market indices (Lee & Kim, 2020). There is a trend to combine DL or LSTM models with other methods such as knowledge graph (Long, Chen, He, Wu, & Ren, 2020), frequency spectra decomposition models (Rezaei et al., 2020), phase-space reconstruction (Yu & Yan, 2020), error correction model (Niu, Wang, Lu, Yang, & Du, 2020) to enhance the overall predictive power. Meanwhile, DL can also be used to develop predictive models based on market sentiment analysis (Li, Bu, Li, & Wu, 2020) and trader emotions (Chun, Ahn, Kim, & Lee, 2020). There is no doubt that with successful stock market predictive models, it would be straightforward to build potential trading strategies (Arévalo et al., 2017; Zhang et al., 2020; Lee & Kim, 2020).

However, deep learning algorithms usually involve a staggering amount of hyper-parameters, especially when they have multiple layers and many neurons. This often makes the training of a layered deep learning algorithm costly and time-consuming. In another approach, the echo state network (ESN) is proposed by utilizing a sparsely connected hidden layer in which training the whole neural network is waived (Jaeger & Haas, 2004). Almost simultaneously, liquid state machines (LSM), which share the same strategy of ESN, are proposed (Maass, Natschläger, & Markram, 2002). ESN and LSM are suitable in time series analysis such as blast furnace gas (Sheng, Zhao, Liu, & Wang, 2012), financial markets (Lin, Yang, & Song, 2009), and object trajectories in robotics (Burgsteiner, Kroll, Leopold, & Steinbauer, 2007). In this line of ESN and LSM studies, a general novel structure called reservoir computing (RC) is introduced (Lukoševičius & Jaeger, 2009; Lukoševičius, Jaeger, & Schrauwen, 2012; Verstraeten, Schrauwen, D'Haene, & Stroobandt, 2007; Appeltant et al., 2011). In a typical RC, a set of nodes in a directed random network serves as the reservoir in which nodes update their states based on the incoming connections and their previous states. Inputs and outputs are connected to some random hidden nodes in the reservoir, respectively. In training, the input vector is projected into a high-dimensional space represented by the reservoir. Instead of training a large number of weights in the reservoir, only output weights are trained using available optimization techniques. This approach dramatically reduces training efforts and speeds up the learning process.

While much-emerging research attention has been focused on RC in time series (Wyffels & Schrauwen, 2010; Butcher, Verstraeten, Schrauwen, Day, & Haycock, 2013), systematic investigations of applying RC in financial market predictions are still insufficient, compared to the rich literature using machine learning or deep learning. Especially, systematic evaluations of reservoirs in different topologies on multiple international markets in substantial time ranges have not been reported yet. To contribute to the study of RC in financial time series prediction, an RC model is proposed in this study. Empirical evaluations have been conducted on seven major international stock markets' daily indices spanning over nine years of 15,723 data points. We systematically investigated three types of network topological structures, namely the Erdős-Rényi random network (ER) (Erdős & Rényi, 1960), Small-World network (SW) (Watts & Strogatz, 1998), and Scale-Free network (SF) (Barabási & Albert, 1999). In comparative experiments, we find that our RC approach outperforms of LSTM (Kim & Won, 2018; Baek & Kim, 2018) and RNN (Rather, Agarwal, & Sastry, 2015; Qin et al., 2017) in predictions of some stock market indices. Additionally, we report that our RC also outperforms two newly proposed deep learning methods reported in recent literature using the same stock market indices datasets (Baek & Kim, 2018; Zhou et al., 2019).

The rest of this paper is organized as follows. First, we give a brief introduction to RC in Section 2. Then we describe our proposed RC model as well as the datasets of seven international major stock markets

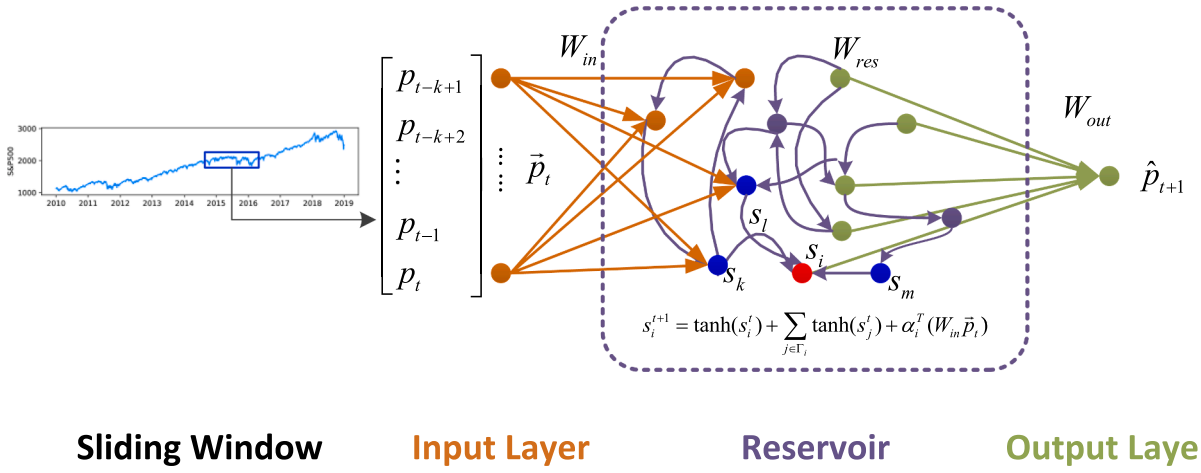


Fig. 2. Structure of the proposed reservoir applied in indices prediction. The prices of each sliding window form an input vector $\vec{p}_t$ and are projected into the reservoir. The internal states of reservoir nodes are updated following the strategy of Eq. 6. The output is readout as the predicted price $\hat{p}_{t+1}$ based on the output layer weight W_{out}.

in Section 3. Afterward, in Section 4, we report the results of systematical experiments using RC models in predictions of the seven market indices. The results of comparative studies to two recently proposed deep learning methods are also reported. We also provide discussions of the impacts of different network topological structures and settings. Finally, Section 5 provides conclusions and discussions of future research.

2. Architecture of reservoir computing

Training the connectivity weight matrices is computationally expensive for most deep learning models (LeCun et al., 2015). Unlike these neural networks in which the weights of all layers are trained, the reservoir computing architecture only requires the readout layer weights to be trained. In contrast, the input network and the reservoir network remain untrained. Initially introduced as Echo State Network (ESN) (Jaeger, 2001) and Liquid State Machines (LSM) (Maass et al., 2002), reservoir computing (RC) models have demonstrated success in a variety of high-dimensional and temporal applications (Schrauwen, Verstraeten, & Campenhout, 2007; Ibáñez-Soria, Garcia-Ojalvo, Soria-Frisch, & Ruffini, 2018; Jaeger & Haas, 2004; Jalalvand, Demuynck, Neve, & Martens, 2018). Thanks to the dynamical nature of a reservoir, RC models have more recently been even implemented as physical reservoirs (Tanaka et al., 2019) using photonic systems (Pauwels, der Sande, Bouwens, Haelterman, & Massar, 2018) or dynamic memristors (Du et al., 2017).

The main structure of an RC model is illustrated in Fig. 1. At time t, the network of nonlinear nodes serves as the reservoir r_t to provide the nonlinear mapping of a temporal L-dimensional input vector x_t into another temporal M-dimensional output vector y_t. Within the reservoir, the internal nodes are connected as a random directed network in which the connectivities are described as a weight matrix $W_{res} \in \mathbb{R}^{N\times N}$, where $N = |r|$ is the size of the reservoir. The elements of W_{res} are randomly initialized and remain unchanged during the training. The input layer L_{in} is a fully connected layer between the input vector x_t and the internal nodes. Similarly, the output layer L_{out} is also a fully connected layer between and the output vector y_t and the internal nodes. L_{in} and L_{out} are described by the weight matrices $W_{in} \in \mathbb{R}^{N\times L}$ and $W_{out} \in \mathbb{R}^{M\times N}$ respectively. Since only some randomly selected input hidden nodes h_{in} in the reservoir are connected to the input vector x_t, only these corresponding elements are non-zero in W_{in}, otherwise zeros for the rest elements. This is the same to W_{res} whereas some randomly selected output hidden nodes h_{out} are connected to the output layer L_{out}. Both W_{in} and W_{res} are randomly initialized and remain unchanged, while W_{out} is trained using techniques like stochastic gradient descent to minimize certain loss functions. This dramatically simplifies the training process, allowing fast learning. The new state vector of the internal reservoir nodes r_{t+1} is excited by the input x_t and the present states r_t. The updating mechanism without considering feedbacks is given as (Schrauwen et al., 2007; Lukoševičius & Jaeger, 2009; Tanaka et al., 2019):

$$r_{t+1} = F(W_{in}x_{t+1} + W_{res}r_t) + b_{in}, \tag{1}$$

where F can be nonlinear activation functions such as $\tanh(\cdot)$ or $\mathrm{ReLU}(\cdot)$, and b_{in} is the input layer bias. For a given node n_i, the state is updated according to its present state s_i^t, the total states of its incoming neighbors $\sum_{j\in\Gamma_i} s_j^t$, and possible contributions from the input $\alpha_i^T(W_{in}x_{t+1})$, namely:

$$s_i^{t+1} = s_i^t + \sum_{j\in\Gamma_i} s_j^t + \alpha_i^T(W_{in}x_{t+1}), \tag{2}$$

where α_i^T is the transposition of a vector with i-th element equals to 1 otherwise 0. From this equation, the reservoir is akin to the Recurrent Neural Network (RNN) (Rumelhart, Hinton, & Williams, 1986) which also utilizes the memories of previous states. This underlying design enables RC to be able to predict time series in scenarios where historical data contains certain levels of information about the evolutions for the systems. The output $y(t)$ is straightforwardly readout as:

$$\hat{y}_{t+1} = W_{out}^t r_t + b_{out}, \tag{3}$$

in which b_{out} is the output layer bias and W_{out}^t is updated iteratively in training stage but unchanged in tests once trainings are finished. The training of W_{out}^t is simply a task of minimizing the loss function of predicted $\hat{y}_{t+1}$ and the ground truth y_{t+1} defined as:

$$\min_{W_{out}^t} \|\hat{y}_{t+1} - y_{t+1}\|^2. \tag{4}$$

Similar to other neural network models, the weight matrices of W_{in}, W_{res}, and W_{out} can be randomly initialized. To allow the reservoir to

Table 1
Seven major stock market indices studied in this work.

Index	Name	Components	Country
S&P 500	S&P 500 Index	500	United States
NYSE	New York Stock Exchange	100	United States
DJI	Dow Jones Industrial Average	30	United States
NASDAQ	Nasdaq Composite Index	2,656	United States
FTSE	Financial Times and Stock Exchange	100	United Kingdom
N225	Nikkei Stock Average	225	Japan
SSE	Shanghai Stock Exchange Component Index	1,239	China

reach steady state, W_{res} can be rescaled as

$$W_{res} := \frac{\rho}{|\lambda_{\max}|} W^0_{res}, \tag{5}$$

where ρ is the spectral radius and $\lambda_{\max}$ stands for the largest eigenvalue of W_{res}, respectively (Jaeger, 2001).

3. Methods and materials

3.1. Methods

In this study, we implement an RC framework to predict stock market indices. The daily closing prices for a given stock market index in the study range form a time series. The whole period is divided into two subsets. The first 80% are used as the training set, while the remaining 20% form the test set. In a sliding window approach, a sequence of price data points $[p_{t-k+1}, p_{t-k+2}, \cdots, p_t]^T$ in k trading days form the input vector $\overrightarrow{p}_t$ to predict the target price p_{t+1}. The predicted output is denoted as $\widehat{p}_{t+1}$. Given the length of a window k and the total length of the time series T, in total we have $T-k+1$ sliding windows. Three types of topological network structures, namely ER, SW, and SF are generated as different RC networks. Two subsets of nodes in RC are randomly selected as the input hidden nodes and output hidden nodes, respectively. The weight matrices W_{in}, and W_{out} are randomly initialized. W_{res} is the adjacent matrix of the reservoir network. For the nodes in the reservoir, we use the following state updating strategy:

$$s_i^{t+1} = \tanh s_i^t + \sum_{j \in \Gamma_i} \tanh s_j^t + \alpha_i^T (W_{in} \overrightarrow{p}_t). \tag{6}$$

Our extensive empirical experiments show that this updating strategy is able to help the loss convergence, which is essential for the W_{out} training in our studies. Fig. 2 provides an overview of the schema of the proposed reservoir structure.

3.2. Data

A stock market index is a benchmark measurement of the whole market performance computed from a portfolio of components which are usually representative stocks from different economic sectors using various weighting methods. Prediction of the movement of a stock market index is important for investment and economic policy-making. In this study, we collect the daily closing prices of seven leading international stock indices, including S&P500 Index (Standard & Poor's), NYSE Composite (New York Stock Exchange), DJI (Dow Jones Industrial Average), Nasdaq Composite Index (Nasdaq Stock Exchange), FTSE 100 Index (Financial Times Stock Exchange 100 Index of London Stock Exchange), Nikkei 225 Index (Tokyo Stock Exchange), and SSE Index (Shanghai Stock Exchange). These are the most followed indices representing major global economies. On the other hand, the prices of these indices are widely used as benchmarks in the literature on predictions of stock indices. Therefore, our study follows this line of literature by using prices of these indices, though some literature use returns instead of prices. The list of indices is presented in Table 1. All prices are historical daily closing prices spanning nine years from January 4, 2000, to December 31, 2018. In total, 15,723 data points are collected from the public financial data services of Yahoo Finance and Investing.org. Table 2 shows the descriptive statistics of all indices. On average, there are over 2,000 trading days for each index. This allows our dataset to be one of the largest datasets used in literature of other machine learning methods as well as RC. The movements of all indices are plotted in Fig. 3. As illustrated, our study period is large enough to contain various market situations. Though the markets are correlated in certain degrees, the predictions of these chaotic markets are conducted as independent tasks. The training and test periods are colored in green and yellow, respectively.

4. Experiments and results

4.1. Experiment settings

As the flowchart illustrated in Fig. 4, the experiment is divided into four stages. First, we collect the daily closing prices of the indices of seven markets from online financial data sources. Afterward, each of the time series is further split into two parts, namely the first 80% trading days form the training set and the remaining 20% form the test set. The former covers over 1,600 trading days, while the latter covers about 400 trading days, respectively. A sliding window approach is adopted to construct each training sample. Different sizes of the window k are considered in this study, namely $k = 5, 10, 15, 20$. In other words, we consider the ranges from a trading week ($k = 5$) to nearly a trading month ($k = 20$). The sliding window is moved forward using the window as the input into the reservoir to predict the following price.

The reservoir might be in different sizes. In this study, three sizes of $N = 10, 50, 100$ are considered. The hidden input nodes h_{in} and hidden output nodes h_{out} are pN randomly selected nodes from the reservoir, we set $p = 0.2, 0.5, 0.7$. Three types of network structures for the reservoir are considered in this study. The ER networks (Erdős & Rényi, 1960) are generated with the edge probability values of $p_{er} = 0.1, 0.2, 0.5$. The expected edge density would be $p_{er}N(N-1)/2$. For SF networks, we adopt the preferential attachment mechanism (Barabási & Albert, 1999) to generate SF networks with combinations of the attaching probability

Table 2
Descriptive statistics of the seven stock market indices over a period of nine years.

Index	Study Period	Trading Days	*Max*	*Min*	*Mean*	*Std.Dev.*
S&P500	01/04/2010–12/28/2018	2,263	2,930.75	1,022.58	1,856.50	520.45
NYSE	12/31/2009–12/28/2018	2,264	13,637.02	6,434.81	9,859.53	1,807.08
DJI	12/31/2009–12/28/2018	2,264	26,828.39	9,686.48	16,624.57	4,447.15
NASDAQ	12/31/2009–12/28/2018	2,264	8,109.69	2,091.79	4,386.49	1,622.66
FTSE	01/04/2010–12/31/2018	2,272	7,877.45	4,805.80	6,429.09	702.92
N225	01/04/2010–12/28/2018	2,208	24,270.62	8,160.01	15,141.53	4,759.24
SSE	01/04/2010–12/28/2018	2,188	5,166.35	1,950.01	2,786.43	555.17

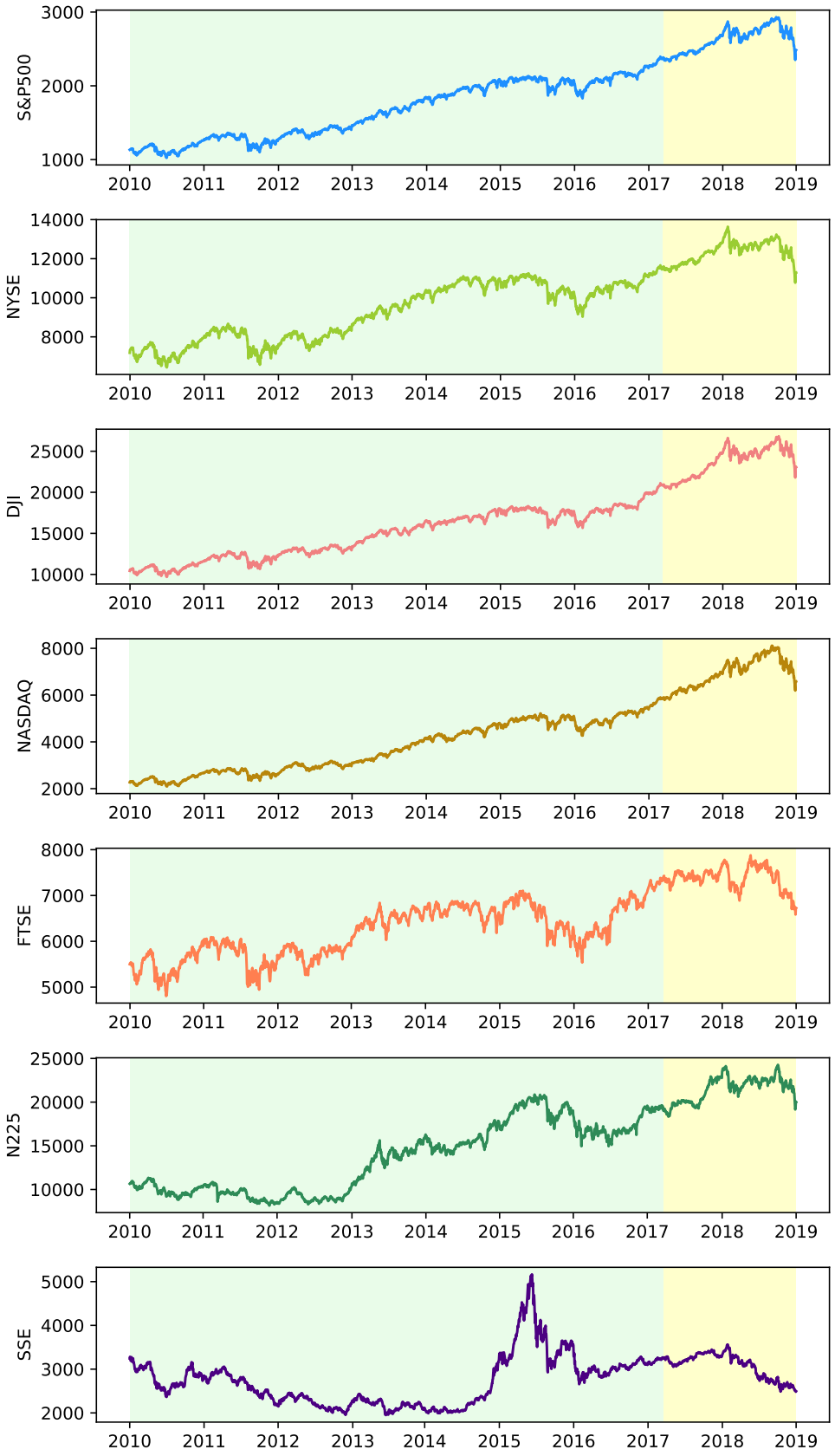


Fig. 3. Plots for the time series of daily closing prices for seven major stock market indices. More than 2000 trading days of nine years between 2010 and 2018 are included in this study. The whole dataset is divided into training and test subsets which are colored in green and yellow respectively.

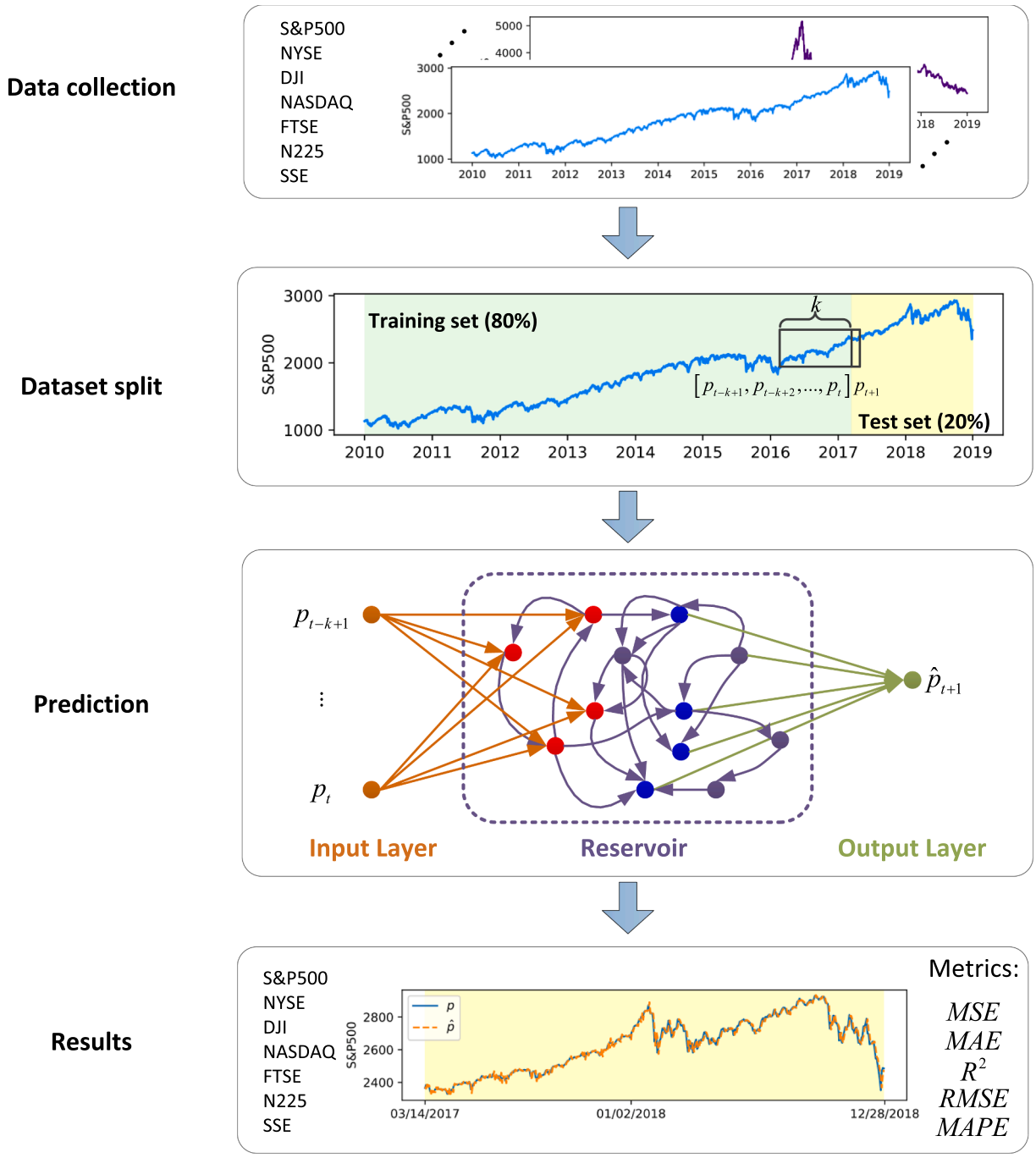


Fig. 4. Flowchart of the proposed RC framework used for stock market indices predictions. First, daily closing prices of seven major market indices are collected; of which the first 80% are used for training and the remaining 20% are used for tests. In predictions, the sliding window approach is utilized to input the prices into the reservoir and a predicted $\widehat{p}_{t+1}$ is readout. The weight matrices W_{out} is trained in the training periods. The performances of the proposed RC models are further evaluated in the test periods and compared with other approaches.

Table 3
Evaluation metrics.

Metric	Description	Definition
MSE	Mean Squared Error	$\frac{1}{T}\sum_{t=1}^{T}(\widehat{p}_t - p_t)^2$
MAE	Mean Absolute Error	$\frac{1}{T}\sum_{t=1}^{T}\lvert\widehat{p}_t - p_t\rvert$
R^2	R-Squared	$1-\frac{\sum_{t=1}^{T}(\widehat{p}_t - p_t)^2}{\widehat{p}_t - \langle p\rangle)^2}$
RMSE	Root Mean Squared Error	$\sqrt{\frac{1}{T}\sum_{t=1}^{T}(\widehat{p}_t - p_t)^2}$
MAPE	Mean Absolute Percentage Error	$\frac{1}{T}\sum_{t=1}^{T}\frac{\lvert\widehat{p}_t - p_t\rvert}{p_t}\times 100\%$

Table 4
Best performances of RC models for the seven major stock market indices.

Index	*MSE*	*MAE*	R^2	*RMSE*	*MAPE*
S&P500	540.74	15.80	0.98	23.25	0.60
NYSE	9241.06	67.96	0.97	96.13	0.55
DJI	50627.75	155.11	0.98	225.01	0.64
NASDAQ	6097.45	53.24	0.98	78.09	0.76
FTSE	2918.69	41.03	0.95	54.02	0.56
N225	52601.34	160.77	0.97	229.35	0.75
SSE	946.35	22.07	0.99	30.76	0.74

$\alpha$and the edge adding probability between two existing nodes β. Both probabilities have values of 0.45, 0.47, and 0.49. For SW networks, the rewiring generative algorithm (Watts & Strogatz, 1998) is adopted with the initial neighbor p_kN, we set $p_k = 0.2, 0.5, 0.7$, and rewiring probability $p_{rewire} = 0.1, 0.2, 0.5$. The weight matrices of input layer W_{in} and output layer W_{out} are both randomly initialized from the uniform distribution $U(-\sqrt{L}, \sqrt{L})$ in the same manner to initialize the forward neural networks. The range for W_{in} is $L = 1/|\vec{p}_t|$, while $L = 1/|h_{out}|$ for W_{out}, respectively. The input and output bias b_{in} are also initialized in the same distribution. The reservoir weight matrix W_{res} is the adjacency matrix initialized in network generating, while the starting internal state vector r^0 is sampled randomly from uniform distribution $U(-0.1, 0.1)$. During training, the output layer weights W_{out} are updated using RMSProp optimizer with learning rates of $a = 0.001, 0.01, 0.1$. For comparative purposes, we also use algorithms of LSTM and RNN to predict the seven stock market indices using the same training and test periods. LSTM and RNN are implemented accordingly to the original models (Hochreiter & Schmidhuber, 1997; Rumelhart et al., 1986). The prediction results of both basis algorithms are compared to our RC models using the same metrics. Since predictions of all market indices are independent tasks, all experiments on the market indices are conducted independently, namely RC models are developed, trained, and tested specifically for each individual index.

4.2. Evaluation metrics

In line with previous stock price prediction literature (Zhang & Wu, 2009; Patel et al., 2015b; Hassan, Nath, & Kirley, 2007), we calculate a bucket of metrics including the mean squared error (*MSE*), mean absolute error (*MAE*), R-squared (R^2), root mean squared error (*RMSE*), and mean absolute percentage error (*MAPE*) to evaluate the performance of our model. The definitions of all the metrics are presented in Table 3. The target price $\widehat{p}_t$ is predicted for each sliding window over the test set before calculating the metrics against the ground truth price p_t. For mean values, the results are averaged over the whole test set.

4.3. Results

In this part, we report the results of two experiments in which the predictions of seven major stock market indices are listed in Section 4.3.1, as well as comparisons to two previous studies (Baek & Kim, 2018; Zhou et al., 2019) in Section 4.3.2, respectively. Finally, we summarize and discuss the influences of RC network topologies and settings on the performances in Section 4.3.3.

4.3.1. Prediction of seven stock market indices

Following the steps as illustrated in the flowchart Fig. 4, we train seven individual RC models for the seven stock market indices. Table 4 presents the results of the best performances of each RC model in the one-day ahead prediction for each index. Values of $MSE, MAE, R^2, RMSE$, and *MAPE* are summarized. From the *MAPE*, we see that the mean absolute percentage errors for all indices are smaller than 1%. This indicates the predictions fit the actual prices closely with small errors. This is further visualized as the plots of time series of predicted price $\widehat{p}_t$ and real price p_t in Fig. 5. The absolute percentage errors (APE) between predicted price $\widehat{p}_t$ and real price p_t are plotted in Fig. 6. As shown, the small values of *APE* indicate the predictive capabilities of RC models for all indices.

To compare the proposed RC models' predictive capability with RNN and LSTM, we implement the RNN and LSTM algorithms and conduct comparative tests using the same test sets. The test periods cover two trading years of more than 400 days between March 14, 2017, to December 31, 2018. As shown in Table 5, each index is predicted using the three algorithms and the *RMSE* and *MAE* are reported. We find that RC outperforms RNN and LSTM in indices of S&P500, NASDAQ, N225, and SSE with smaller *RMSE*. According to *MAE*, RC outperforms RNN and LSTM in indices of NASDAQ, N225, and SSE, while for the rest of indices, RNN has a slightly smaller *MAE*. For simplicity, we only report the detailed results of *RMSE* and *MAE* for the three algorithms in Table 5. However, we also calculate measurements of $R^2, RMSE$, and *MAPE* for the three algorithms. For these three measurements, RC achieves the best performances in three indices (NASDAQ, FTSE, and SSE) except S&P500, NYSE, DJI, and N225, for which RNN has the best performances. The results show that RC can achieve competitive performances of the state-of-the-art deep learning methods of LSTM and RNN.

4.3.2. Comparative study

To further compare our proposed RC models' predictive capabilities with the models reported in recent studies, a comparative experiment is conducted in which we compare with two separate studies of applying deep learning algorithms in predictions of market indices (Zhou et al., 2019; Baek & Kim, 2018). In the first study (Zhou et al., 2019), a neural network model EMD2FNN is proposed in predictions of NASDAQ index and SSE index in a study period of 1,214 trading days between January 4, 2012, and December 30, 2016, and S&P500 index in a study period of 1,260 trading days between January 3, 2007, and December 30, 2011. In another study (Baek & Kim, 2018), a two-parts LSTM model ModAugNet is proposed in predictions of S&P500 index and KOSPI200 (Korea Stock Exchange KOSPI 200 Index) in a study period of more than 4,300 trading days between January 4, 2000, and July 27, 2017. These newly proposed deep learning models achieve competitive prediction performances (Zhou et al., 2019; Baek & Kim, 2018). In our comparative experiment, we first develop RC models following the strategy as previously described. To allow direct comparisons, we train and test RC models using the same study periods and calculate the same metrics of *MAE*, *RMSE*, *MAPE*, and *MSE* as adopted in the two studies.

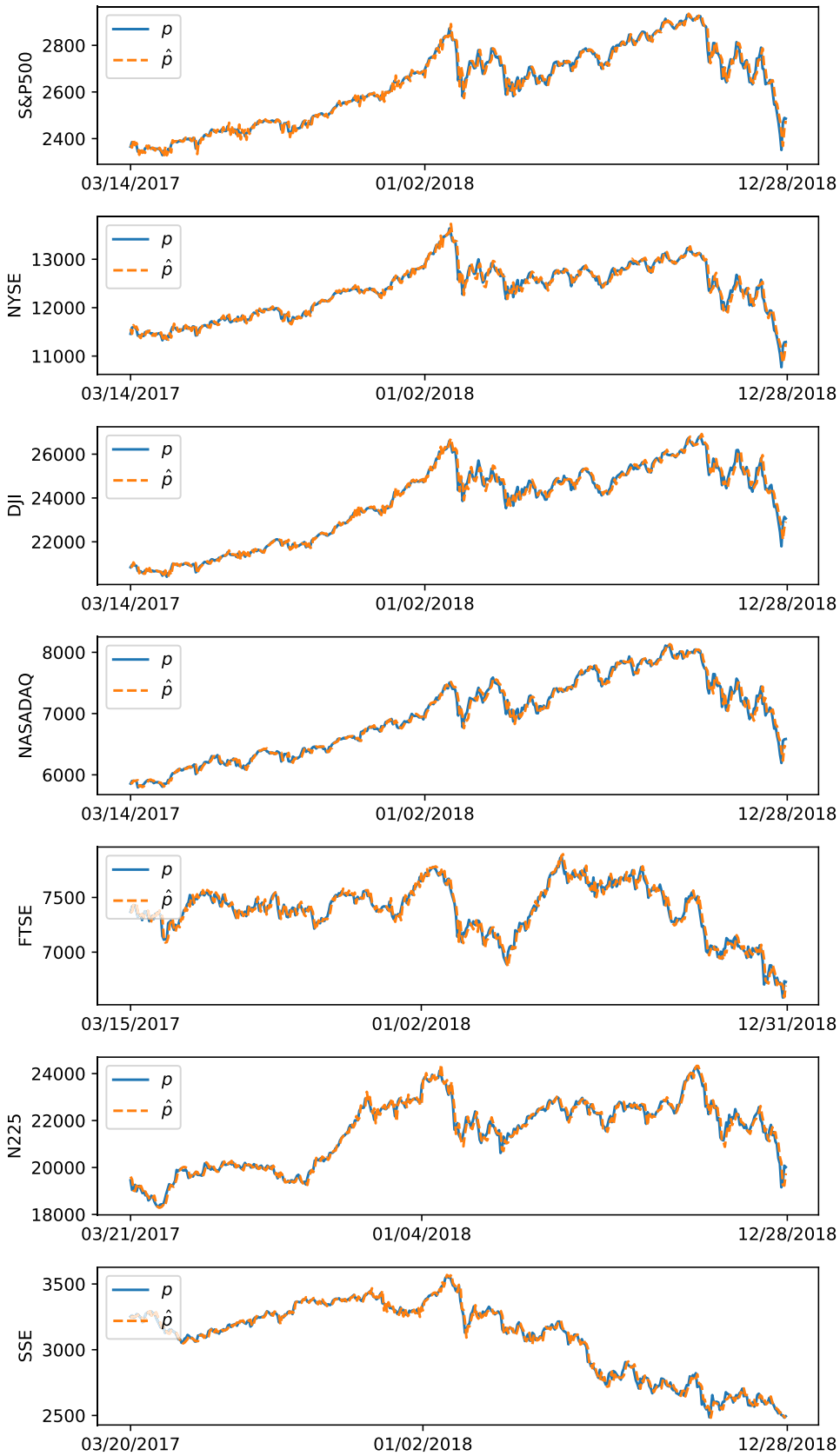


Fig. 5. Plots of predicted index price $\widehat{p}_t$ and ground truth price p_t for each stock market index in the test period. The predicted prices are results of respective best performing RC models obtained in the training period. As illustrated, the predicted curves closely follow the movements of ground truth curves.

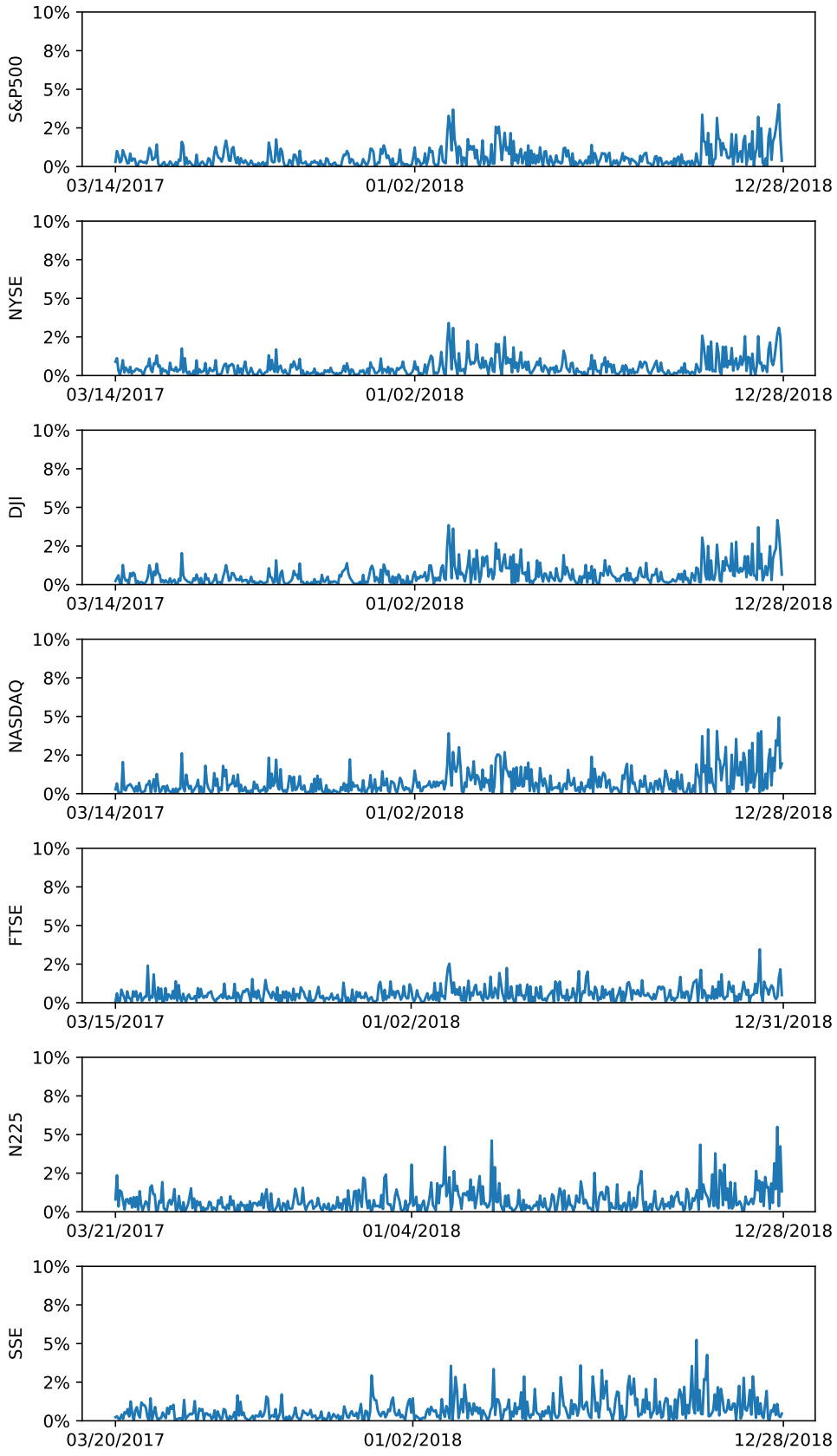


Fig. 6. Absolute percentage error (*APE*) for each stock market index in the test period.

Table 5
Comparisons of the *RMSE* and *MAE* for the models of proposed RC, RNN, and LSTM. The experiments are conducted using the same test periods for all indices. The bolded values are the best results achieved by the best performing corresponding model for each index.

Index	Test Days	RMSE			MAE		
		RC	RNN	LSTM	RC	RNN	LSTM
S&P500	453	**23.25**	23.44	23.89	15.80	**15.37**	15.66
NYSE	453	96.13	**94.49**	110.82	67.96	**63.99**	76.04
DJI	453	225.01	**221.97**	256.78	155.11	**146.21**	173.52
NASDAQ	453	**78.09**	78.80	81.41	**53.24**	53.80	56.11
FTSE	454	54.02	**53.70**	57.24	41.03	**40.62**	43.11
N225	442	**229.35**	229.92	267.24	**160.77**	163.33	190.58
SSE	438	**30.76**	30.89	33.08	**22.07**	22.08	23.99

Table 6
Performance comparisons of the reported results of EMD2FNN (Zhou et al., 2019) and our RC model for three stock market indices of S&P500, NASDAQ, and SSE. Bolded values indicate better results. The same trading periods are used to form the test sets for S&P500 (January 3, 2011 to December 30, 2011), NASDAQ (January 4, 2016 to December 30, 2016), and SSE (January 5, 2016 to December 30, 2016), respectively.

Index	Test days	EMD2FNN (Zhou et al., 2019)			Proposed RC		
		MAE	*RMSE*	*MAPE*	*MAE*	*RMSE*	*MAPE*
S&P500	252	**13.04**	**17.66**	**1.05**	13.59	18.00	1.10
NASDAQ	251	52.48	70.46	1.08	**35.90**	**49.23**	**0.73**
SSE	243	37.18	61.51	1.23	**28.53**	**42.72**	**0.96**

Table 7
Performance comparisons of the reported results of ModAugNet (Baek & Kim, 2018) and our RC model for two stock market indices of S&P500 and KOSPI200. Bolded values indicate better results. The trading period between January 2, 2008 to July 27, 2017 is used as the test set for S&P500 and KOSPI200.

Index	Test days	ModAugNet (Baek & Kim, 2018)			Proposed RC		
		MSE	*MAPE*	*MAE*	*MSE*	*MAPE*	*MAE*
S&P500	2409	342.48	1.08	**12.06**	**285.54**	**0.87**	12.07
KOSPI200	2376	**7.56**	1.01	**1.98**	7.85	**0.90**	2.02

Firstly, as shown in Table 6, RC outperforms the EMD2FNN (Zhou et al., 2019) in predictions of NASDAQ index and SSE index, while slightly underperforms in the prediction of S&P500 index, though the results are very close to EMD2FNN (Zhou et al., 2019). Specifically, in the predictions of NASDAQ and SSE, RC achieves significant 31.59% and 23.27% decreases of *MAE*, 30.13% and 30.55% decreases of *RMSE*, 32.41% and 21.95% decreases of *MAPE*, respectively. Secondly, as the results shown in Table 7, RC outperforms ModAugNet (Baek & Kim, 2018) in prediction of S&P500 index while slightly underperforms in prediction of KOSPI200. In the prediction of S&P500, RC achieves 16.63% and 19.44% decreases of *MSE* and *MAPE*, respectively. These comparative investigations into recent results achieved by using EMD2FNN (Zhou et al., 2019) and ModAugNet (Baek & Kim, 2018) demonstrate that RC is also a promising model in stock market index prediction with competitive predictive capabilities.

4.3.3. *Impacts of RC networks and settings*

To investigate how the topological structure impacts the performance of a reservoir model, we generated reservoirs in three types of networks, namely random network (ER), scale-free network (SF), and small-world network (SW). As shown in Table 8, the best performing structure is SF for predictions of DJI, FTSE, and SSE, SW for NYSE, NASDAQ and N225, and ER for S&P500, respectively. These networks are also in different sizes ranging from 10 to 100. This shows that the best performances could be achieved for different index predictions by different structures with different sizes. No single structure with a certain size can perform well in all cases. Furthermore, as illustrated in Fig. 7, the three types of structures perform close to each other in the prediction of the same index. However, the metrics reported in the Table 8 show that there are still significant differences for different combinations of structures, sizes, lengths of input windows, and learning rates in the training of output weights. The best performances are achieved when the learning rate is 0.1 in predictions of S&P500, while the learning rate is 0.001 for the rest indices. Larger learning rates tend to speed up the learning process but might not even converge. From the results, we see that lower learning rates are acceptable, since our main objective is to achieve best predictions. Different sizes of input windows are evaluated to see what is the optimal input size of historical trading data for each index. The results show that RC achieves best performances when the input is set to an optimal length, which is 5 for NASDAQ, 15 for S&P500, and 20 for NYSE, DJI, FTSE, N225, and SSE, respectively. In the financial industry, the five trading days is a calendar week, 15 trading days is about half of calendar month, and 20 trading days is roughly a calendar month. Our extensive experiments hint that a large input window of 20 trading days is sufficient, and small input windows might not increase the performances in most predictions. The results also tell us that a systematic investigation of different combinations of determinant factors, including the topological structures, sizes, learning rates, and lengths of input windows, is necessary to achieve good predictions.

5. Conclusion

Applying innovative machine learning techniques to predict movements of financial markets using price time series has important implications for the financial industry (Bose & Mahapatra, 2001). More recently, machine learning and deep learning models such as LSTM and RNN have been investigated in various stock markets (Selvin, Vinayakumar, Gopalakrishnan, Menon, & Soman, 2017). Reservoir computing (RC) is proposed as a special RNN model. A set of randomly connected state nodes provides the nonlinearity to project the inputs into a high-dimensional space in the model. Instead of training all weights in traditional methods, RC only requires the output layer weights to be trained while keeping the input layer's weights and the reservoir network fixed. This approach significantly simplifies training. Thanks to the nonlinear memory nature of the reservoir network, RC models have been applied in predictions of time series (Butcher et al., 2013; Wyffels & Schrauwen, 2010; Ferreira & Ludermir, 2008; Escalona-Morán,

Table 8
Best performances for individual index achieved by different network structures and settings. Three types of networks are random network (ER), scale-free network (SF), small-world network (SW). The reservoir size, length of input data, and learning rate are denoted as N, k, and α respectively.

Index	Structure	N	k	α	*MSE*	*MAE*	R^2	*RMSE*	*MAPE*
S&P500	ER	50	15	0.100	540.74	15.80	0.98	23.25	0.60
	SF	100	20	0.001	573.05	16.33	0.98	23.94	0.62
	SW	50	20	0.001	563.14	16.22	0.98	23.73	0.62
NYSE	ER	100	20	0.001	9500.15	69.05	0.97	97.47	0.56
	SF	100	20	0.001	9504.31	70.48	0.97	97.49	0.57
	SW	50	20	0.001	9241.06	67.96	0.97	96.13	0.55
DJI	ER	100	20	0.001	54612.58	157.39	0.98	233.69	0.65
	SF	100	20	0.001	50627.75	155.11	0.98	225.01	0.64
	SW	100	20	0.001	53547.44	155.93	0.98	231.40	0.65
NASDAQ	ER	100	20	0.001	6220.89	54.57	0.98	78.87	0.78
	SF	50	20	0.001	6333.27	56.05	0.98	79.58	0.80
	SW	100	5	0.001	6097.45	53.24	0.98	78.09	0.76
FTSE	ER	100	20	0.001	2939.43	42.07	0.95	54.22	0.57
	SF	100	20	0.001	2918.69	41.03	0.95	54.02	0.56
	SW	50	10	0.001	2950.25	41.87	0.95	54.32	0.57
N225	ER	100	20	0.001	52934.53	164.47	0.97	230.08	0.76
	SF	50	15	0.001	52842.90	166.78	0.97	229.88	0.77
	SW	100	20	0.001	52601.34	160.77	0.97	229.35	0.75
SSE	ER	50	10	0.001	988.89	22.56	0.99	31.45	0.75
	SF	50	20	0.001	946.35	22.07	0.99	30.76	0.74
	SW	10	5	0.001	952.26	22.13	0.99	30.86	0.74

Soriano, García-Prieto, Fischer, & Mirasso, 2014). As a novel model, applications of RC in financial markets are still far less explored compared to other well-studied machine learning or deep learning models such as LSTM and RNN.

In this work, we develop RC models using the random network, small-world network, and scale-free network and systematically investigate the predictive capabilities in one-day ahead predictions of seven international major stock market indices with comparisons to LSTM and RNN. The seven market indices of S&P500 Index, NYSE Composite, DJI, Nasdaq Composite Index, FTSE 100 Index, Nikkei 225 Index, and SSE Index are representatives of the most followed global financial markets. We collect the time series of daily closing prices between January 4, 2010, and December 28, 2018, covering 2000 trading days for each index. We train and test the models over the first 80% (1600 days) and over the rest 20% (400 days), respectively. Metrics of *MSE*, *MAE*, R^2, *RMSE*, and *MAPE* are evaluated for each model. We find that RC outperforms LSTM and RNN in six indices according to *RMSE* and two indices according to *MAE*. In another comparative experiment, RC outperforms EMD2FNN (Zhou et al., 2019) in predicting NASDAQ and SSE, and outperforms ModAugNet (Baek & Kim, 2018) in the prediction of S&P500. Overall, the experiments demonstrate that RC can achieve competitive performances in most cases compared to the state-of-the-art models of LSTM and RNN.

However, this study still has limitations that need further studies. Firstly, the dataset only covers seven major stock market indices. Though this study unitizes a larger dataset with more indices and larger time ranges compared to other studies, it would be worth doing empirical tests that consider the time series of more individual stocks from different markets. Meanwhile, readers should be noted that these indices are correlated, and using a limited number of indices would hinder the generalization of predictive models. Though the indices are predicted independently, adding more indices into the study would significantly improve the diversity, reduce the correlations among indices, and lead to more reliable models with better generalization. Secondly, the present dataset only considers daily closing prices. One might be interested in investigating how RC performs in other data frequencies of time-scales such as hour, minute, second, or even ultra-high frequencies. Thirdly, this study only focuses on stock markets. However, it would be natural to expand the current research to other financial markets such as bond markets, commodity markets, money markets, derivatives markets, futures markets, foreign exchange markets, and interbank markets. Fourthly, considering that the objective of the present study is to investigate the RC models, no practical trading strategies are developed based on the results achieved by RC. In further studies, we might build trading strategies based on predictions of RC models and further evaluate the values of the models in backtesting using real financial time series. Lastly, the proposed method is only applied to the time series of closing prices. It would be interesting to further evaluate the proposed RC models in other price time series of open, high, low, and market time series such as returns and risks. Furthermore, there is a trend of combining different machine learning or deep learning methods together in market predictions (Cervelló-Royo & Guijarro, 2020; Long et al., 2020; Rezaei et al., 2020; Yu & Yan, 2020; Niu et al., 2020). However, it's largely unexplored to combine RC with existing machine learning or deep learning methods. This ensemble approach would be exciting and promising for future research lines.

The prediction of financial markets is so challenging that it's hard for a single method to achieve the best performances in predictions of all individual indices. As shown by our results, RC outperforms other methods in many cases but still can not dominate in all predictions of all indices. However, this does not undermine the potential value of RC in predictions of financial markets. A practitioner is supposed to evaluate all possible methods before putting them into applications wisely. We believe that RC is a promising candidate adding to the present machine learning toolbox of financial market analysis. Hopefully, these promising research directions would open wide, rich possibilities and fruitful study topics for scholars. These further studies of RC in predictions of financial time series will also inspire new tools and applications for financial industry practitioners.

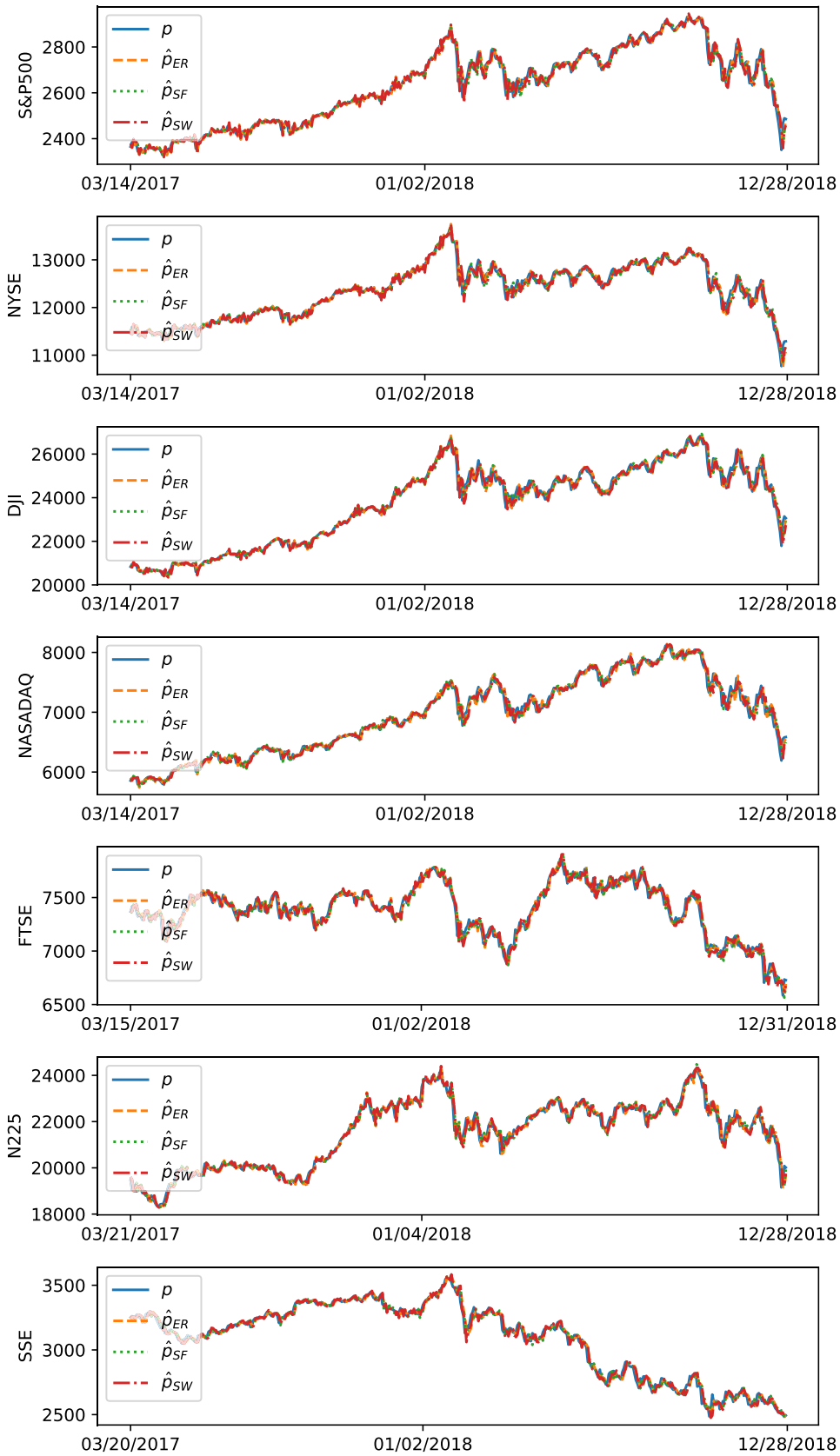


Fig. 7. Plots of predicted curves obtained using different reservoir network topologies and the ground truth curves. All of the three topologies of random network (ER), scale-free network (SF), and small-world network (SW) are able to follow the movements of ground truth curves.

CRediT authorship contribution statement

Wei-Jia Wang: Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing - original draft. **Yong Tang:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Supervision, Visualization, Validation, Writing - original draft, Writing - review & editing. **Jason Xiong:** Formal analysis, Investigation, Project administration, Validation, Writing - original draft, Writing - review & editing. **Yi-Cheng Zhang:** Investigation, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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